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INTRODUCTION

Retail sales datasets usually contain thousands of transactions recorded across products, stores and time periods. Although most transactions follow a common pattern, some values are unusually high or low. These observations are called outliers. They may indicate genuine events such as bulk purchases, promotional campaigns or seasonal demand, or they may reveal errors in data entry, pricing or quantity. Exploratory Data Analysis (EDA) helps identify these observations through statistics and graphs before they influence business conclusions.

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OBJECTIVE AND IMPORTANCE

OBJECTIVES

- Identify unusual sales observations.
- Understand possible causes of extreme values.
- Apply IQR and Z-score methods.
- Visualize outliers using charts.
- Improve the quality of business analysis.

WHY IT IS IMPORTANT

Outlier detection is important because unusual sales values can strongly affect the mean, variance, graphs, forecasting and business decisions. It helps identify data-entry errors, duplicate transactions, incorrect prices and quantities, while also revealing genuine events such as bulk purchases, seasonal demand and promotional sales. Detecting these values improves data quality, makes sales patterns easier to understand, supports better inventory planning and helps identify possible anomalies or suspicious transactions.

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DATASET AND VARIABLES

A typical retail sales dataset contains transaction-level information that allows analysts to examine sales from multiple perspectives such as product, store and time.

Product / Category Item purchased	Quantity Units sold	Unit Price Price per unit	Sales Amount Quantity × Price
Date Transaction day	Store Branch location	Customer Segment Buyer group	

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DATA PREPARATION

Before detecting outliers, the data should be cleaned so that unusual values caused by poor data quality are not confused with genuine business events.

Data Preparation Task	What to Check	Purpose
Check Missing Values	Empty quantity, price, date or category	Prevent incomplete records from affecting analysis
Remove Duplicates	Repeated transaction IDs or identical records	Avoid counting the same sale more than once
Validate Ranges	Quantity, price and sales amount	Identify impossible or suspicious values
Standardize Data Types	Numbers, dates and categories	Ensure calculations and comparisons work correctly
Calculate Sales Amount	Quantity × Unit Price	Create a consistent sales measure

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OUTLIER DETECTION METHODS

IQR METHOD

Q1 = First Quartile
Q3 = Third Quartile
IQR = Q3 – Q1
Lower Bound = Q1 – 1.5 × IQR
Upper Bound = Q3 + 1.5 × IQR

Values outside the bounds are flagged for investigation.

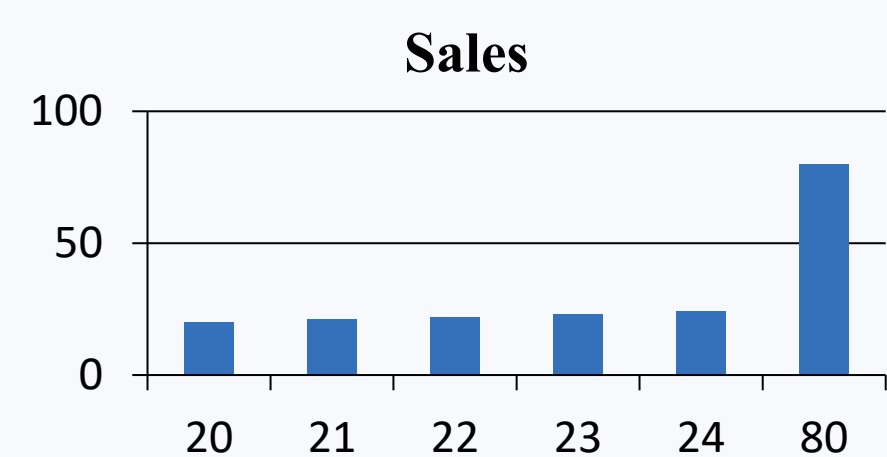
Z-SCORE METHOD

$z = (x - \mu) / \sigma$

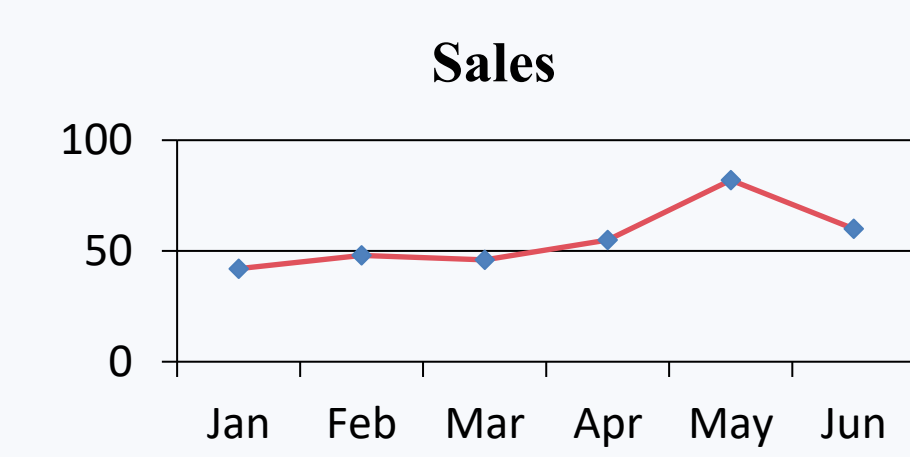
The Z-score method measures how far a sales value is from the mean in terms of standard deviations and is calculated using $z = (x - \mu) / \sigma$, where **x** is the observation, **μ** is the mean and **σ** is the standard deviation. A Z-score close to zero indicates that the value is near the average, while a large positive or negative Z-score indicates that the value is far from the mean and may be an outlier. The method is useful for standardizing observations and identifying unusually high or low sales values.

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EXPLORATORY DATA ANALYSIS



Sales distribution with an extreme value



Monthly sales trend showing a spike

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KEY FINDINGS AND BUSINESS BENEFITS

KEY FINDINGS

- Extreme values may reveal large transactions.
- Some outliers are caused by data-entry errors.
- Promotions and seasonal demand can create genuine spikes.
- Graphs make unusual patterns easier to understand.
- Statistical rules should be combined with business context.

BUSINESS BENEFITS

- Better sales forecasting.
- Smarter inventory planning.
- Faster anomaly investigation.
- More reliable reports.
- Better support for management decisions.

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CONCLUSION

Outlier analysis is an important part of exploratory data analysis. The IQR and Z-score methods provide simple statistical ways to identify unusual observations, while graphs such as bar charts and trend plots help communicate their effect clearly. The best approach is to detect, visualize and investigate outliers before deciding whether they should be corrected, retained or documented.